Screening Dyslexia in Cantonese-Speaking Children with Machine Learning Approaches: Testing Auditory Processing Indicators

Developmental dyslexia is a neurodevelopmental disorder that impairs reading and writing acquisition and is typically diagnosed only after children begin to struggle with literacy [1]. Although children with dyslexia may improve following remediation [2], early screening and detection in pre-readers is essential to maximize the benefits of early intervention. Phonological deficits are widely considered the core impairment in dyslexia, and these deficits may originate from more fundamental auditory processing difficulties that can be assessed before children become literate. However, it remains debated whether the auditory deficits contributing to dyslexia are universal across languages or language-specific. The Temporal Sampling Framework (TSF) proposes that impaired discrimination of amplitude (assessed through rise time discrimination) is associated with dyslexia across languages [3], whereas the Language-Specific Auditory Cues (LSAC) hypothesis suggests that the auditory cues supporting literacy development vary by language, with pitch discrimination being especially important in tonal languages such as Chinese [4]. In addition to linguistic and sensory deficits, cognitive deficits (e.g. working memory, attention and inhibition) have also been reported in dyslexia [5].

The current study is a first step towards developing early screening tools for pre-literate children, by applying machine learning models to classify Cantonese-speaking primary school children with and without dyslexia. We focused on three theoretically or empirically motivated factors: phonological awareness, auditory threshold, and cognitive functions. Specifically, we aimed to address three questions: (1) whether phonological awareness, auditory thresholds, and cognitive functions can classify Cantonese children with dyslexia using machine learning methods; (2) which factor is most important for predicting dyslexia; and (3) the relative importance of pitch-based versus time-based auditory processing skills in Cantonese reading.

Thirty children with dyslexia (ages 7–9) and thirty age-matched controls were assessed on phonological awareness (segmental and suprasegmental), auditory discrimination (rise time, pitch height, and pitch glide), and cognitive abilities (nonverbal IQ, working memory, and inhibition). Seven commonly used machine learning models were applied to predict dyslexia status: Naive Bayes, Support Vector Machine (RBF), K-Nearest Neighbors, Decision Tree, Artificial Neural Network, Logistic Regression, and Random Forest.

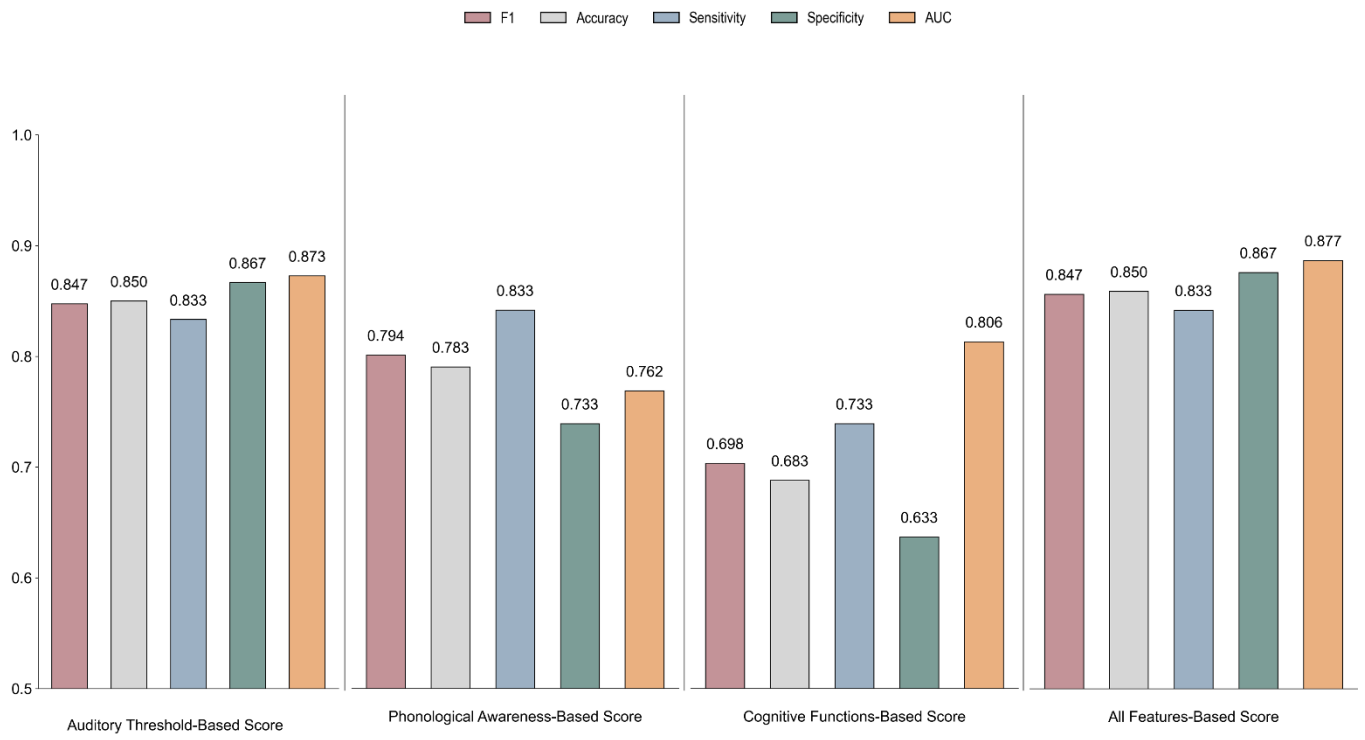
Among these models, the Random Forest model achieved the highest F1 score, accuracy, and AUC (Table 1) and was therefore selected for further analysis. Feature importance analysis revealed that auditory thresholds were the most influential factor (normalized importance: 0.72), followed by cognitive functions (0.23), and phonological awareness (0.04). Notably, when the three factor sets were entered separately into the Random Forest model, the auditory-threshold subset alone achieved F1, accuracy, and AUC scores comparable to those of the full model containing all features (Figure 1). Further analysis of the auditory-threshold subset showed that pitch height discrimination (0.54) was the most important feature, followed by pitch glide (0.28) and rise time (0.18), underscoring the critical role of pitch processing in Cantonese dyslexia screening.

These results suggest that phonological awareness, auditory thresholds, and cognitive functions—particularly auditory processing measures—have the potential to serve as early indicators of dyslexia risk even before literacy acquisition begins, thereby maximizing opportunities for early intervention. Moreover, our findings highlight the importance of pitch discrimination in identifying Cantonese-speaking children with dyslexia. Future research should further validate the predictive value of auditory measures through longitudinal designs.

Table 1. Algorithms Performance of All Models.

Model	ROC-AUC	Accuracy	F1 Score	Precision	Sensitivity	Specificity
GaussianNB	0.83	0.75	0.73	0.80	0.67	0.83
SVM (RBF)	0.83	0.82	0.83	0.77	0.90	0.73
KNN (k=5)	0.77	0.70	0.69	0.71	0.67	0.73
Decision Tree	0.73	0.73	0.74	0.72	0.77	0.70
ANN (3-layer)	0.83	0.77	0.77	0.75	0.80	0.73
Logistic Regression	0.83	0.78	0.79	0.77	0.80	0.77
Random Forest	**0.88**	**0.85**	**0.85**	**0.86**	**0.83**	**0.87**

Figure 1. Classification performance of the Random Forest on different data sets.



Reference

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